

AERIS Expert Review Packet

Anomaly

Source / metric	tceq / no2
Observed / expected	18.2 / 3.06768
Time	2026-06-16 00:00 UTC
Location	29.6863, -95.2947
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	humidity	percent	9 / 616	57.73 to 100; mean 89.1584	94.24 at 2026-06-15 23:55Z; dt 5 min
asos	temperature	degC	9 / 616	22.7778 to 33.3333; mean 26.7453	26 at 2026-06-15 23:55Z; dt 5 min
asos	wind_direction	degree	9 / 616	0 to 360; mean 120.211	20 at 2026-06-15 23:55Z; dt 5 min
asos	wind_speed	m/s	9 / 616	0 to 11.3178; mean 2.5889	2.05778 at 2026-06-15 23:55Z; dt 5 min
noaa_gfs	gh_500	m	11 / 143	5851.27 to 5926.57; mean 5888.9	5885.77 at 2026-06-16 00:00Z; dt 0 min
noaa_gfs	pbl_height	m	11 / 143	32.0322 to 1802.53; mean 342.235	48.199 at 2026-06-16 00:00Z; dt 0 min
noaa_gfs	precipitable_water	mm	11 / 143	52.1939 to 69.5435; mean 59.9716	60.9815 at 2026-06-16 00:00Z; dt 0 min
noaa_gfs	surface_pressure	hPa	11 / 143	1001.26 to 1014.43; mean 1008.97	1009.44 at 2026-06-16 00:00Z; dt 0 min
noaa_gfs	t_850	degC	11 / 143	16.785 to 19.498; mean 18.322	18.8384 at 2026-06-16 00:00Z; dt 0 min
noaa_gfs	u_10m	m/s	11 / 143	-4.71304 to 1.58078; mean -0.9417	-0.619447 at 2026-06-16 00:00Z; dt 0 min
noaa_gfs	v_10m	m/s	11 / 143	-1.42555 to 6.14875; mean 1.9996	-0.575546 at 2026-06-16 00:00Z; dt 0 min
openaq	ozone	ppm	17 / 1110	-0.001 to 0.031; mean 0.0135	0.01 at 2026-06-16 00:00Z; dt 0 min
openaq	pm10	ug/m3	3 / 212	5 to 110; mean 25.0377	14 at 2026-06-16 00:00Z; dt 0 min
openaq	pm25	ug/m3	90 / 3111	-2.6 to 52; mean 6.9223	5.7 at 2026-06-16 00:00Z; dt 0 min
openweather	cloud_cover	percent	5 / 355	3 to 100; mean 91.6113	100 at 2026-06-16 00:01Z; dt 1.8 min
openweather	humidity	percent	5 / 355	68 to 100; mean 90.5859	88 at 2026-06-16 00:01Z; dt 1.8 min
openweather	precipitation	mm/h	5 / 355	0 to 24.39; mean 1.1861	0.1 at 2026-06-16 00:01Z; dt 1.8 min
openweather	pressure	hPa	5 / 355	1005 to 1016; mean 1011.44	1011 at 2026-06-16 00:01Z; dt 1.8 min
openweather	temperature	degC	5 / 355	24.22 to 33.3; mean 26.8152	26.72 at 2026-06-16 00:01Z; dt 1.8 min
openweather	wind_direction	degree	5 / 355	0 to 358; mean 159.786	89 at 2026-06-16 00:01Z; dt 1.8 min
openweather	wind_speed	m/s	5 / 355	0.07 to 5.67; mean 2.0365	2.07 at 2026-06-16 00:01Z; dt 1.8 min
purpleair	pm25	ug/m3	66 / 4758	0 to 46.2; mean 8.1691	8.7 at 2026-06-16 00:00Z; dt 0 min

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
sentinel5p	s5p_aer_ai_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-06-15 19:09Z; dt 290.4 min
sentinel5p	s5p_ch4_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-06-15 19:09Z; dt 290.4 min
sentinel5p	s5p_co_column	mol/m^2	1 / 1	0.0259566 to 0.0259566; mean 0.026	0.0259566 at 2026-06-14 19:24Z; dt 1715.4 min
sentinel5p	s5p_co_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-06-15 19:09Z; dt 290.4 min
sentinel5p	s5p_hcho_column	mol/m^2	1 / 1	0.000111617 to 0.000111617; mean 0.0001	0.000111617 at 2026-06-14 19:24Z; dt 1715.4 min
sentinel5p	s5p_hcho_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-06-15 19:09Z; dt 290.4 min
sentinel5p	s5p_no2_column	mol/m^2	1 / 1	2.03572e-05 to 2.03572e-05; mean 0	2.03572e-05 at 2026-06-14 19:24Z; dt 1715.4 min
sentinel5p	s5p_no2_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-06-15 19:09Z; dt 290.4 min
sentinel5p	s5p_o3_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-06-15 19:09Z; dt 290.4 min
sentinel5p	s5p_so2_column	mol/m^2	1 / 1	-6.10873e-06 to -6.10873e-06; mean -0	-6.10873e-06 at 2026-06-14 19:24Z; dt 1715.4 min
sentinel5p	s5p_so2_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-06-15 19:09Z; dt 290.4 min
tceq	co	ppm	3 / 213	0.1 to 1.1; mean 0.223	0.3 at 2026-06-16 00:00Z; dt 0 min
tceq	no2	ppb	12 / 790	-2.1 to 27; mean 5.6039	18.2 at 2026-06-16 00:00Z; dt 0 min
tceq	so2	ppb	3 / 219	-0.6 to 0.6; mean -0.1169	-0.5 at 2026-06-16 00:00Z; dt 0 min

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	humidity	06-14 12Z 80.89, 18Z 69.98, 06-15 00Z 88.16, 06Z 92.9, 12Z 93.04, 18Z 90.65, 06-16 00Z 93.26, 06Z 97.13, 12Z 93.15, 18Z 86.17, 06-17 00Z 89.65, 06Z 95.39
asos	temperature	06-14 12Z 28.76, 18Z 30.74, 06-15 00Z 27.65, 06Z 26.9, 12Z 25.98, 18Z 26.07, 06-16 00Z 25.96, 06Z 25.46, 12Z 25.49, 18Z 26.98, 06-17 00Z 26.19, 06Z 24.8
asos	wind_direction	06-14 12Z 147, 18Z 176.3, 06-15 00Z 115, 06Z 124, 12Z 173.5, 18Z 106.5, 06-16 00Z 68.46, 06Z 67.17, 12Z 155.8, 18Z 106.6, 06-17 00Z 106, 06Z 101.3
asos	wind_speed	06-14 12Z 3.611, 18Z 4.066, 06-15 00Z 2.229, 06Z 2.379, 12Z 2.744, 18Z 1.271, 06-16 00Z 1.533, 06Z 1.126, 12Z 3.255, 18Z 2.048, 06-17 00Z 2.815, 06Z 3.921
noaa_gfs	gh_500	06-14 12Z 5911, 18Z 5926, 06-15 00Z 5921, 06Z 5922, 12Z 5903, 18Z 5893, 06-16 00Z 5886, 06Z 5891, 12Z 5872, 18Z 5865, 06-17 00Z 5855, 06Z 5858, 12Z 5853
noaa_gfs	pbl_height	06-14 12Z 257.7, 18Z 1416, 06-15 00Z 468, 06Z 225.2, 12Z 264.2, 18Z 316.4, 06-16 00Z 147.4, 06Z 74.46, 12Z 157.9, 18Z 420.5, 06-17 00Z 286.1, 06Z 131.4, 12Z 283.8
noaa_gfs	precipitable_water	06-14 12Z 57.96, 18Z 59.03, 06-15 00Z 60.59, 06Z 59.99, 12Z 58.11, 18Z 64.91, 06-16 00Z 61.25, 06Z 62.82, 12Z 64.55, 18Z 63.2, 06-17 00Z 55.72, 06Z 56.3, 12Z 55.2
noaa_gfs	surface_pressure	06-14 12Z 1012, 18Z 1012, 06-15 00Z 1010, 06Z 1012, 12Z 1011, 18Z 1011, 06-16 00Z 1009, 06Z 1009, 12Z 1008, 18Z 1008, 06-17 00Z 1005, 06Z 1005, 12Z 1003
noaa_gfs	t_850	06-14 12Z 18.52, 18Z 18.04, 06-15 00Z 18.96, 06Z 19.25, 12Z 18.61, 18Z 17.21, 06-16 00Z 18.8, 06Z 18.41, 12Z 17.78, 18Z 17.53, 06-17 00Z 18.87, 06Z 18.04, 12Z 18.19
noaa_gfs	u_10m	06-14 12Z -1.28, 18Z -0.8351, 06-15 00Z -1.073, 06Z -0.4866, 12Z -0.3975, 18Z -1.02, 06-16 00Z -0.3913, 06Z -0.6458, 12Z 0.4544, 18Z 0.4365, 06-17 00Z -1.765, 06Z -2.342, 12Z -2.896

Source	Metric	Values
noaa_gfs	v_10m	06-14 12Z 2.038, 18Z 4.411, 06-15 00Z 3.057, 06Z 2.524, 12Z 2.016, 18Z 3.091, 06-16 00Z -0.2301, 06Z 1.278, 12Z 1.881, 18Z 4.033, 06-17 00Z 0.5866, 06Z 0.9076, 12Z 0.4017
openaq	ozone	06-14 12Z 0.01561, 18Z 0.02208, 06-15 00Z 0.01393, 06Z 0.01061, 12Z 0.008989, 18Z 0.01508, 06-16 00Z 0.007862, 06Z 0.002926, 12Z 0.01025, 18Z 0.01699, 06-17 00Z 0.01466, 06Z 0.02004, 12Z 0.01982
openaq	pm10	06-14 12Z 33, 18Z 22.17, 06-15 00Z 33.06, 06Z 49.44, 12Z 41.22, 18Z 12.22, 06-16 00Z 20.89, 06Z 23.69, 12Z 18.33, 18Z 15, 06-17 00Z 18.89, 06Z 14.44, 12Z 13.67
openaq	pm25	06-14 12Z 7.408, 18Z 8.402, 06-15 00Z 9.636, 06Z 9.642, 12Z 7.469, 18Z 4.56, 06-16 00Z 7.97, 06Z 7.125, 12Z 4.252, 18Z 2.61, 06-17 00Z 2.806, 06Z 4.772, 12Z 4.544
openweather	cloud_cover	06-14 12Z 45.19, 18Z 81.5, 06-15 00Z 95.83, 06Z 99.86, 12Z 99.13, 18Z 99.93, 06-16 00Z 99.57, 06Z 99.33, 12Z 99.9, 18Z 100, 06-17 00Z 96.81, 06Z 75.69
openweather	humidity	06-14 12Z 84.92, 18Z 73.03, 06-15 00Z 89, 06Z 92.72, 12Z 93.48, 18Z 94.13, 06-16 00Z 92.93, 06Z 95.23, 12Z 94.17, 18Z 91.79, 06-17 00Z 91.1, 06Z 93.86
openweather	precipitation	06-14 12Z 0.1485, 18Z 0.1917, 06-15 00Z 0.203, 06Z 0.19, 12Z 3.307, 18Z 4.336, 06-16 00Z 0.7973, 06Z 1.585, 12Z 1.259, 18Z 0.7517, 06-17 00Z 0.4219, 06Z 0.7976
openweather	pressure	06-14 12Z 1015, 18Z 1013, 06-15 00Z 1013, 06Z 1013, 12Z 1014, 18Z 1012, 06-16 00Z 1012, 06Z 1011, 12Z 1011, 18Z 1009, 06-17 00Z 1008, 06Z 1006
openweather	temperature	06-14 12Z 28.83, 18Z 31.85, 06-15 00Z 27.99, 06Z 27.09, 12Z 26.29, 18Z 25.43, 06-16 00Z 25.95, 06Z 25.36, 12Z 25.52, 18Z 26.41, 06-17 00Z 26.26, 06Z 25.04
openweather	wind_direction	06-14 12Z 170, 18Z 182, 06-15 00Z 151.6, 06Z 164.7, 12Z 187.4, 18Z 163.7, 06-16 00Z 125.5, 06Z 198.5, 12Z 184.4, 18Z 165, 06-17 00Z 114.1, 06Z 111.2
openweather	wind_speed	06-14 12Z 2.405, 18Z 3.247, 06-15 00Z 2.106, 06Z 2.097, 12Z 1.986, 18Z 1.815, 06-16 00Z 1.037, 06Z 1.571, 12Z 1.688, 18Z 2.229, 06-17 00Z 2.293, 06Z 2.014
purpleair	pm25	06-14 12Z 8.86, 18Z 10.99, 06-15 00Z 11.83, 06Z 10.83, 12Z 7.916, 18Z 6.546, 06-16 00Z 9.512, 06Z 9.382, 12Z 4.645, 18Z 5.254, 06-17 00Z 4.878, 06Z 7.361, 12Z 7.826
sentinel5p	s5p_aer_ai_granule_available	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	s5p_ch4_granule_available	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	s5p_co_granule_available	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	s5p_hcho_granule_available	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	s5p_no2_granule_available	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	s5p_o3_granule_available	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	s5p_so2_granule_available	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
tceq	co	06-14 12Z 0.2111, 18Z 0.1933, 06-15 00Z 0.2167, 06Z 0.1778, 12Z 0.2333, 18Z 0.2667, 06-16 00Z 0.2778, 06Z 0.2278, 12Z 0.2556, 18Z 0.2444, 06-17 00Z 0.2167, 06Z 0.1556, 12Z 0.2
tceq	no2	06-14 12Z 2.338, 18Z 1.518, 06-15 00Z 2.643, 06Z 2.819, 12Z 5.514, 18Z 7.59, 06-16 00Z 11.11, 06Z 8.731, 12Z 5.738, 18Z 7.23, 06-17 00Z 6.751, 06Z 4.889, 12Z 4.05
tceq	so2	06-14 12Z -0.2333, 18Z -0.1278, 06-15 00Z -0.1833, 06Z -0.2389, 12Z -0.2111, 18Z -0.07222, 06-16 00Z 0.01667, 06Z -0.09444, 12Z -0.08333, 18Z -0.07222, 06-17 00Z 0.005556, 06Z -0.1, 12Z -0.1667

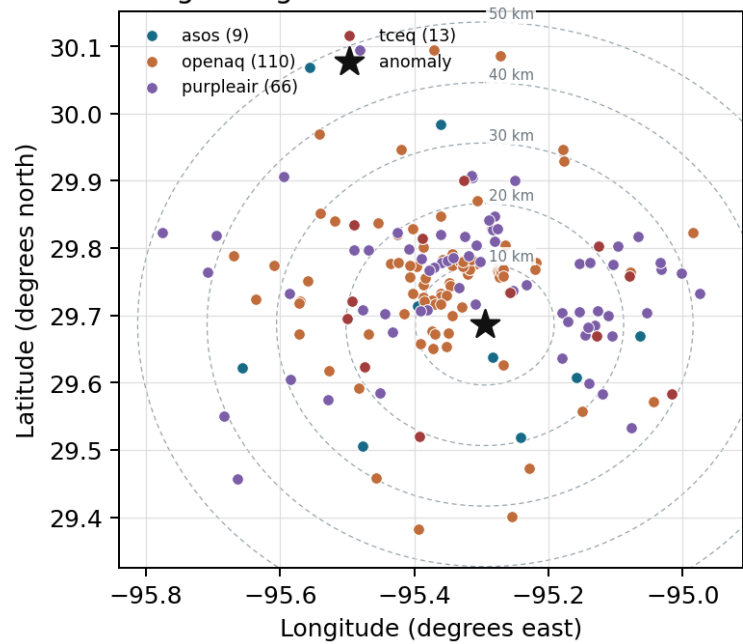
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

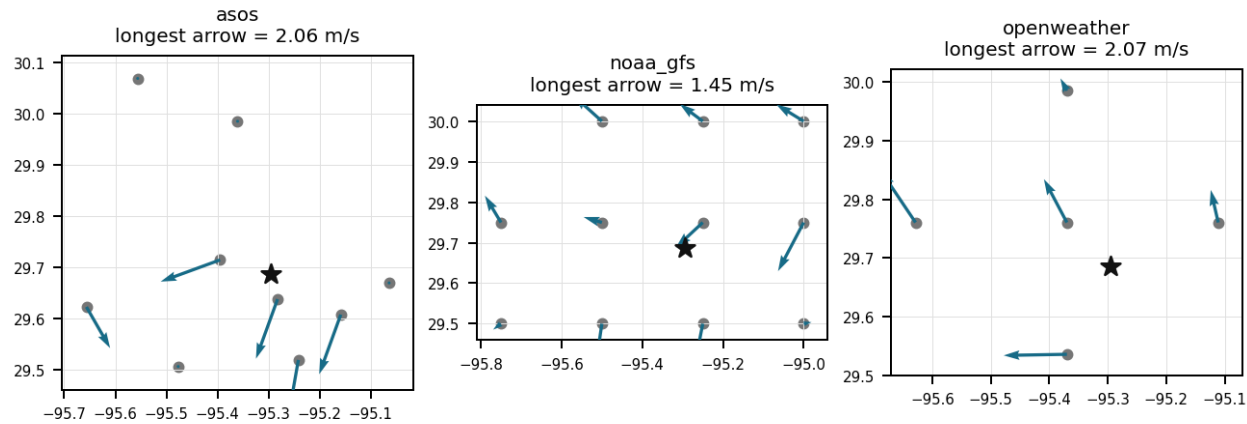
Source	Metric	Values
asos	humidity	HOU (5.555 km) 87.63, MCJ (10.163 km) 92.6, EFD (15.811 km) 87.54, LVJ (19.307 km) 87.54, T41 (22.362 km) 87.58, AXH (26.678 km) 86.21, IAH (33.753 km) 91.33, SGR (35.675 km) 92.16, +1 more
asos	temperature	HOU (5.555 km) 27.12, MCJ (10.163 km) 26.33, EFD (15.811 km) 27.25, LVJ (19.307 km) 26.96, T41 (22.362 km) 27.2, AXH (26.678 km) 26.59, IAH (33.753 km) 26.48, SGR (35.675 km) 26.54, +1 more
asos	wind_direction	HOU (5.555 km) 136.8, MCJ (10.163 km) 125.7, EFD (15.811 km) 105.5, LVJ (19.307 km) 122.6, T41 (22.362 km) 147.9, AXH (26.678 km) 62.36, IAH (33.753 km) 135, SGR (35.675 km) 126.8, +1 more
asos	wind_speed	HOU (5.555 km) 3.008, MCJ (10.163 km) 3.701, EFD (15.811 km) 3.318, LVJ (19.307 km) 2.315, T41 (22.362 km) 3.43, AXH (26.678 km) 1.079, IAH (33.753 km) 2.351, SGR (35.675 km) 2.701, +1 more
noaa_gfs	gh_500	gfs:29.75,-95.25 (8.297 km) 5888, gfs:29.75,-95.50 (21.05 km) 5889, gfs:29.50,-95.25 (21.162 km) 5889, gfs:29.50,-95.50 (28.689 km) 5890, gfs:29.75,-95.00 (29.33 km) 5888, gfs:30.00,-95.25 (35.148 km) 5888, gfs:29.50,-95.00 (35.231 km) 5889, gfs:30.00,-95.50 (40.109 km) 5888, +3 more
noaa_gfs	pbl_height	gfs:29.75,-95.25 (8.297 km) 352.1, gfs:29.75,-95.50 (21.05 km) 373.5, gfs:29.50,-95.25 (21.162 km) 335.1, gfs:29.50,-95.50 (28.689 km) 303.2, gfs:29.75,-95.00 (29.33 km) 353.5, gfs:30.00,-95.25 (35.148 km) 357.2, gfs:29.50,-95.00 (35.231 km) 437.7, gfs:30.00,-95.50 (40.109 km) 398.2, +3 more
noaa_gfs	precipitable_water	gfs:29.75,-95.25 (8.297 km) 60.38, gfs:29.75,-95.50 (21.05 km) 59.7, gfs:29.50,-95.25 (21.162 km) 60.39, gfs:29.50,-95.50 (28.689 km) 60.32, gfs:29.75,-95.00 (29.33 km) 60.51, gfs:30.00,-95.25 (35.148 km) 59.67, gfs:29.50,-95.00 (35.231 km) 60.17, gfs:30.00,-95.50 (40.109 km) 59.07, +3 more
noaa_gfs	surface_pressure	gfs:29.75,-95.25 (8.297 km) 1010, gfs:29.75,-95.50 (21.05 km) 1008, gfs:29.50,-95.25 (21.162 km) 1010, gfs:29.50,-95.50 (28.689 km) 1009, gfs:29.75,-95.00 (29.33 km) 1011, gfs:30.00,-95.25 (35.148 km) 1008, gfs:29.50,-95.00 (35.231 km) 1011, gfs:30.00,-95.50 (40.109 km) 1007, +3 more
noaa_gfs	t_850	gfs:29.75,-95.25 (8.297 km) 18.3, gfs:29.75,-95.50 (21.05 km) 18.27, gfs:29.50,-95.25 (21.162 km) 18.38, gfs:29.50,-95.50 (28.689 km) 18.29, gfs:29.75,-95.00 (29.33 km) 18.32, gfs:30.00,-95.25 (35.148 km) 18.27, gfs:29.50,-95.00 (35.231 km) 18.49, gfs:30.00,-95.50 (40.109 km) 18.27, +3 more
noaa_gfs	u_10m	gfs:29.75,-95.25 (8.297 km) -0.7707, gfs:29.75,-95.50 (21.05 km) -1.022, gfs:29.50,-95.25 (21.162 km) -1.205, gfs:29.50,-95.50 (28.689 km) -1.28, gfs:29.75,-95.00 (29.33 km) -0.7837, gfs:30.00,-95.25 (35.148 km) -0.7484, gfs:29.50,-95.00 (35.231 km) -0.9445, gfs:30.00,-95.50 (40.109 km) -0.773, +3 more
noaa_gfs	v_10m	gfs:29.75,-95.25 (8.297 km) 1.655, gfs:29.75,-95.50 (21.05 km) 1.644, gfs:29.50,-95.25 (21.162 km) 2.206, gfs:29.50,-95.50 (28.689 km) 1.846, gfs:29.75,-95.00 (29.33 km) 2.177, gfs:30.00,-95.25 (35.148 km) 1.8, gfs:29.50,-95.00 (35.231 km) 3.429, gfs:30.00,-95.50 (40.109 km) 1.781, +3 more
openaq	ozone	325 (0.01 km) 0.01311, 3378 (6.399 km) 0.009662, 320 (7.255 km) 0.01229, 2039 (11.575 km) 0.01153, 1319333 (13.71 km) 0.01526, 2046 (15.8 km) 0.009925, 332 (16.16 km) 0.01793, 312 (18.679 km) 0.0142, +9 more
openaq	pm10	13380082 (6.399 km) 27.21, 15852419 (9.465 km) 27.4, 271 (16.16 km) 20.43
openaq	pm25	14140751 (0.011 km) 6.825, 15990898 (4.432 km) 5.912, 16564662 (4.852 km) 6.304, 13418566 (5.486 km) 6.499, 8967123 (6.375 km) 9.868, 8967479 (6.375 km) 8.683, 15539682 (6.38 km) 8.647, 3379 (6.399 km) 8.614, +82 more
openweather	cloud_cover	grid:center (10.975 km) 91.99, grid:south (18.24 km) 92.2, grid:east (19.553 km) 92.3, grid:west (33.267 km) 93.52, grid:north (33.994 km) 88.06
openweather	humidity	grid:center (10.975 km) 91.8, grid:south (18.24 km) 88.17, grid:east (19.553 km) 90.61, grid:west (33.267 km) 91.1, grid:north (33.994 km) 91.25
openweather	precipitation	grid:center (10.975 km) 1.009, grid:south (18.24 km) 1.33, grid:east (19.553 km) 1.405, grid:west (33.267 km) 1.124, grid:north (33.994 km) 1.062
openweather	pressure	grid:center (10.975 km) 1011, grid:south (18.24 km) 1011, grid:east (19.553 km) 1012, grid:west (33.267 km) 1011, grid:north (33.994 km) 1012
openweather	temperature	grid:center (10.975 km) 26.9, grid:south (18.24 km) 26.91, grid:east (19.553 km) 27.05, grid:west (33.267 km) 26.57, grid:north (33.994 km) 26.65
openweather	wind_direction	grid:center (10.975 km) 158, grid:south (18.24 km) 155.3, grid:east (19.553 km) 166.2, grid:west (33.267 km) 158.8, grid:north (33.994 km) 160.6
openweather	wind_speed	grid:center (10.975 km) 1.534, grid:south (18.24 km) 2.546, grid:east (19.553 km) 2.595, grid:west (33.267 km) 1.496, grid:north (33.994 km) 2.011

Source	Metric	Values
purpleair	pm25	6752 (3.679 km) 8.015, 99797 (6.962 km) 7.184, 133994 (7.18 km) 0.8767, 293725 (8.647 km) 10.35, 166435 (8.846 km) 10.44, 293735 (9.495 km) 10.54, 222601 (10.473 km) 5.223, 235425 (11.32 km) 8.752, +58 more
tceq	co	482011035 (6.378 km) 0.1352, 482011039 (16.162 km) 0.1333, 482011052 (16.841 km) 0.4043
tceq	no2	482010416 (0.0 km) 5.662, 482011035 (6.378 km) 8.662, 482011039 (16.162 km) 5.049, 482011052 (16.841 km) 14.91, 482011066 (19.515 km) 7.518, 482010055 (19.781 km) 5.327, 480391004 (20.723 km) 2.266, 482010026 (20.843 km) 7.172, +4 more
tceq	so2	482011035 (6.378 km) -0.4192, 482011039 (16.162 km) 0.02466, 482010051 (18.688 km) 0.04384

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Claim 1 of 36

Concentrated primary nitrogen oxide emissions from evening traffic and industrial sources accumulated locally within a shallow boundary layer.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 2 of 36

TCEQ ground monitors recorded a severe NO₂ concentration anomaly of 18.2 ppb at 2026-06-16T00:00:00+00:00 in Houston relative to an expected baseline of 3.07 ppb.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 3 of 36

Coinciding with the NO₂ peak at 2026-06-16T00:00:00Z, surface ozone measured across 17 OpenAQ sensors dropped to a 6-hour average of 0.00786 ppm.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 4 of 36

Ground-based monitoring confirmed a broad regional rise in 6-hour mean NO₂ concentrations up to 11.11 ppb, centered on the anomaly monitor's 18.2 ppb peak.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 5 of 36

A sharp drop in planetary boundary layer height accompanied by weak surface wind speeds inhibited vertical and horizontal dispersion, trapping pollutants near the surface.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 6 of 36

On June 16, 2026 at 00:00 UTC, a severe NO₂ concentration anomaly of 18.2 ppb was recorded by a TCEQ monitor at coordinates (29.686, -95.295), significantly exceeding the expected baseline of 3.07 ppb with a z-score of 7.21.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 7 of 36

A nearby fresh combustion plume from Houston urban-roadway, industrial, marine, rail, or other fuel-burning activity is a likely direct cause of the severe NO₂ enhancement because the anomalous monitor measured 18.2 ppb while area TCEQ NO₂ was elevated at the same time and contemporaneous ozone at the collocated OpenAQ location was low at about 0.01 ppm, which is consistent with fresh NO_x influence near emission sources.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 8 of 36

The anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter into the atmosphere.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 9 of 36

The location of the anomaly is in Houston, Texas, with a grid center point at (10.975 km) 91.8 and a distance of approximately 3.679 km from sensor 6752.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 10 of 36

The TCEQ Houston monitor at 29.686, -95.295 recorded a severe NO₂ anomaly of 18.2 ppb at 2026-06-16 00:00 UTC against an expected value near 3.07 ppb, and the surrounding TCEQ network also showed elevated 6-hour mean NO₂ at 00Z relative to earlier periods.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 11 of 36

Weak dispersion is a likely contributing cause of the severe NO₂ enhancement because winds near the event were light at roughly 1 to 2 m/s and the modeled planetary boundary layer height was very shallow near the anomaly time, reaching about 48 m at the nearest GFS grid cell, which would favor trapping of primary NO_x near the surface.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 12 of 36

The poor-dispersion accumulation hypothesis is strongly supported because meteorology near 2026-06-16T00:00:00Z featured very high humidity around 94 percent, light surface winds near 1 to 2 m/s, nearly complete cloud cover near 100 percent, and a very shallow modeled planetary boundary layer near 48 m, all of which favor trapping of NO₂ close to sources.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 13 of 36

The pollutant anomaly is a TCEQ NO2 measurement of 18.2 ppb at 2026-06-16T00:00:00+00:00 at 29.686, -95.295 in Houston, Texas.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 14 of 36

The air quality anomaly in Houston, Texas is caused by high levels of NO2 particles.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 15 of 36

The chemical pattern near 2026-06-16 00:00 UTC is most consistent with a fresh local NOx emission plume because ozone was low while PM2.5, PM10, CO, and SO2 did not show comparably strong coincident spikes and satellite context does not indicate a broad regional pollution enhancement at the anomaly time.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 16 of 36

The air quality anomaly is related to particulate matter (PM2.5) with a value of 8.7 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 17 of 36

The air quality anomaly in Houston, Texas is characterized by elevated levels of PM2.5 and NO2.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 18 of 36

The combination of high PM2.5 and NO2 levels is a significant contributor to the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 19 of 36

Moist, cloudy, nighttime conditions are a likely contributing cause of the severe NO2 enhancement because relative humidity was around 90 to 95 percent, cloud cover was near 100 percent, and ozone was depressed during the event, a combination that favors reduced photolysis and nocturnal chemical partitioning that can maintain emitted NOx as NO2 close to sources.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 20 of 36

The anomalous TCEQ NO2 value of 18.2 ppb is far above the expected value of 3.0676829268292685 ppb and has a z-score of 7.210558990276541, which is labeled severe in the anomaly record.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 21 of 36

ASOS weather stations and OpenWeather observations recorded stagnant surface wind speeds below 1.6 m/s and cloud cover near 100% at the time of the anomaly.

Mark exactly one:☐ V☐ I☐ U**Note:**

Claim 22 of 36

The air quality anomaly in Houston, Texas is caused by high levels of CO particles.

Mark exactly one:☐ V☐ I☐ U**Note:**

Claim 23 of 36

During the 6-hour period around 2026-06-16T00:00:00Z, the regional average NO2 concentration across 12 TCEQ monitors reached a period peak of 11.11 ppb.

Mark exactly one:☐ V☐ I☐ U**Note:**

Claim 24 of 36

A severe thunderstorm or tornado event occurred in the area, stirring up pollutants and reducing air quality.

Mark exactly one:☐ V☐ I☐ U**Note:**

Claim 25 of 36

The anomaly is a result of a combination of factors including high temperatures, low humidity, and stagnant atmospheric conditions, leading to poor air mixing and increased pollutant concentrations.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 26 of 36

Planetary boundary layer heights dropping below 150 meters combined with light surface wind speeds near 1.5 m/s around 2026-06-16T00:00:00Z severely suppressed atmospheric dispersion.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 27 of 36

Cloud cover near 100% attenuated solar radiation, weakening photolysis rates and suppressing surface ozone formation while maintaining high ground-level NO₂ concentrations.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 28 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM_{2.5} particles.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 29 of 36

The anomaly occurred on June 16th, 2026, at approximately 00:00:00 UTC.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 30 of 36

NOAA GFS model output showed a collapse of the planetary boundary layer height down to 147.4 meters around 2026-06-16T00:00:00+00:00, severely restricting vertical atmospheric mixing.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 31 of 36

At 2026-06-16T00:00:00+00:00, nearby observations show openaq ozone of 0.01 ppm at the anomaly location and a 6-hour mean TCEQ NO2 of 11.11 ppb for 06-16 00Z, which is higher than the earlier 6-hour means of 7.59 ppb at 06-15 18Z and 5.514 ppb at 06-15 12Z.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 32 of 36

The broad regional or elevated-column enhancement hypothesis is weakened because particulate matter remained moderate near the anomaly, sulfur dioxide showed no concurrent increase, and the available Sentinel-5P trace-gas columns are either temporally distant from the event or do not show corroborating evidence for a large regional pollution episode.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 33 of 36

Houston-area meteorology at 2026-06-16 00:00 UTC was highly favorable for near-surface NO_x accumulation because wind speeds were light, boundary-layer height was very low, humidity was high, and cloud cover was nearly complete.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 34 of 36

Overcast cloud cover reduced solar radiation and photolysis rates of nitrogen dioxide, enhancing its accumulation near the ground.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 35 of 36

The local fresh-combustion plume hypothesis is supported because the anomalous TCEQ monitor measured NO₂ at 18.2 ppb versus an expected 3.07 ppb at 2026-06-16T00:00:00Z, while nearby ozone was only 0.01 ppm and nearby CO was modestly elevated to 0.3 ppm, a combination consistent with recently emitted NO_x near the surface rather than a well-mixed aged regional plume.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 36 of 36

Industrial activities and vehicle traffic are likely contributing factors to the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.
